# SECTION 16725

# ACCESS CONTROL SYSTEMS

## PART 1 GENERAL

### 1.1 INTRODUCTION

A. This section includes product information and installation requirements for work done on the access control system. This includes complete new systems in new buildings as well as modifications to the existing access control system.

### 1.2 DESCRIPTION

A. Furnish and install a complete and operable security system as described herein. This work shall include providing the necessary labor, services and material as required by the security system manufacturer. to provide a complete and functional security system.

B. The Contractor Shall:

C. Furnish and install security system panels, equipment, and devices as detailed on the drawings.

1). For each access control point shown on the Drawings:

2). Furnish and install all applicable devices where indicated on drawings.

3). Furnish and install new cables to connect all field devices such as; card readers, electric locking devices, request to exit devices, magnetic switches, audible sounders, and key switches to the associated security panel.

4). Install the latest firmware as specified by the manufacturer.

5). Complete the system programming and database entry.

6). Test cables installed under this section for shorts or opens as described under PART 3.

7). Label cables as described under PART 3.

8). Complete the functional acceptance tests.

9). Furnish and install all conduit, raceways, and fittings as described under part 2.8.

## 1.3 INCORPORATED DOCUMENTS

A. Sections 16010 Electrical General Provisions, 16050 Basic Electrical Materials & Methods, and 16130 Raceways and Boxes apply to all electrical work in this Section.

B. The following related work will be performed under other Sections:

1). Installation of electrical 120 volt branch circuits to the access control system Division 16 Electrical.

2). Installation of magnetic door switches, magnetic locks, and electric strikes - Section 08700 FINISH HARDWARE.

3). Cut-outs for the magnetic door switches, electric strikes, electric hinges, and power transfers - Section 08100 METAL DOORS AND FRAMES.

4). Tie in to the fire alarm system panel - Section 16721 FIRE ALARM SYSTEM.

5). Installation of category 6A network communication cable to the access control system panel - Section 16740.

C. Micron Security Systems Installation & Maintenance Specification

## 1.4 SYSTEM DESCRIPTION AND FUNCTION

A. The access control system shall be incorporated into the central access control data base so as to become an integral part of the plant-wide access control system with all inherent functions and capabilities. The access control system shall include the following subsystems.

1). Access Control

2). Intrusion Detection

## 1.5 SUBMITTALS

A. Before executing any work, submit for approval product data and shop drawings as specified in Section 01300 (in a completely marked and coordinated package) to assure full compliance with the contract requirements. Information shall be submitted for all material and equipment the access control contractor proposes to furnish to accomplish the contract work. Submittals for each manufactured item shall contain descriptive literature (equipment specification), equipment drawings, diagrams, and catalog cuts of the proposed equipment. The submittal shall also include manufacturer names, catalog models and numbers, layout dimensions and capacities. Drawings shall indicate adequate clearance for operation, maintenance, and replacement of operating equipment and devices. All submittals to be approved by owner prior to commencing work.

B. The following operational and maintenance material shall be submitted at the completion of the work as required in Section 01700:

1). Bound books, with index and index tabs, to include the following for all products and devices furnished under this Specification.

2). Hardware Manuals

3). Programming Manuals

4). Maintenance Manuals

5). Hardware Diagnostics

C. Parts lists, availability (supplier name, telephone number, and location), and guarantee of local stock for all products and devices furnished under this Specification.

D. List of recommended spare equipment, along with quantities the Owner should maintain on site.

E. Final approved set of all shop drawing submittals.

1.6 INSTALLATION AND START-UP REQUIREMENTS

A. Installation

1). Owner will specify locations of all field devices prior to installation by the access control contractor.

2). Provide a written punch list of items remaining to be completed to Owner 2 weeks prior to start-up date.

3). The new equipment shall be incorporated in the existing LENEL computer database by Owner.

B. Start-up

1). Start-up and testing of the system will be done by the access control contractor with the assistance of the Owner. During start-up, the access control contractor will be responsible for repairing any mistakes in wiring or installation immediately. As a minimum, one access control system technician familiar with the installation shall be available at all time during start-up for the repairs.

## 1.7 AS-BUILT DOCUMENTATION

A. The access control contractor shall keep one print of each drawing at their shop and in the field. See also Section 01010. These prints are to be marked up as work progresses. The access control contractor will provide as-built drawings at the end of this job. The as-built drawings shall be created using the marked up prints as reference. A construction set of the floor plan with devices, equipment, and notes will be given to the access control contractor. The access control contractor shall be required to draw in the as-built conduit, wirefill, color coding and miscellaneous changes to create “as built” drawings.

B. Revised description of operation for configuration and up-to-date I/O data base reflecting the “as-built” condition of the system at the time of final acceptance by Owner.

## 1.8 GUARANTEE

A. Guarantee all materials and workmanship for a period of one year after acceptance of the building by the Owner. The access control contractor shall make all necessary repairs, adjustments, and replacements at no cost to the Owner during this period. Provide a written letter of warranty to Owner on all equipment and services, including system configuration, supplied by the access control contractor outlining start-up dates, duration of warranty and terms of warranty. Refer to Section 01700 for additional requirements.

# PART 2 PRODUCTS

## 2.1 GENERAL REQUIREMENTS

A. The access control contractor shall furnish all equipment required for a complete installation which is not supplied by the Owner or other contractors as part of the requirements listed in other Sections of this Specification. In general, the scope of work included in Part 1 of this Section describes equipment furnished by the Owner or others. Where the access control contractor is responsible for installing equipment furnished by the Owner or others, he will take full responsibility for the care and access control of that equipment upon receipt of the equipment at the job site.

B. All access control contractor-furnished equipment shall be new and left in their original container until installed. The Owner reserves the right to inspect this equipment prior to installation and reject any equipment not cared for in an appropriate or defined fashion

## PART 3 EXECUTION

### 3.1 GENERAL

A. The access control system shall be connected to perform all functions and operate according to the system description and function contained in Part 1. The access control contractor shall coordinate with other contractors working in the same areas or providing support services (120 volt AC, network) to his equipment

### 3.2 EQUIPMENT INSTALLATION AND LABELING REQUIREMENTS

A. Equipment shall be installed in accordance with the drawings and specifications and in strict accordance with the manufacturer’s recommendations. All equipment, which must be serviced, operated or maintained, shall be located in fully accessible locations. Locations shown on the drawings are diagrammatic but approximate the ideal location for the equipment at the time of design. Changes to these locations shall be coordinated with the Owner.

B. All electrical equipment must be located so as to provide minimum working space as required by the NEC.

C. The access control contractor shall coordinate with the Architect as to the exact location of all devices, wire-way and conduit locations prior to the installation.

D. For all unattended entry points, (turnstiles, vehicle gates, etc.) a camera will be required and two (2) phones (internal/external). The LENEL 1320 panel shall be mounted locally.

### 3.3 DEVICE INSTALLATION AND LABELING REQUIREMENT

A. Labeling of devices is required at the panel and at the field device. Circuit numbers and labeling will be provided by Micron Security during design/construction.

### 3.4 CONDUIT/WIRE INSTALLATION AND LABELING

A. All wire and cable shall be continuous from terminal to device with no splices. Splice at device is acceptable with butt splices only other methods such as. Telephone splices are strictly prohibited.

B. All cables and wire shall have wire labels which shall be Brady Datab type or equivalent approved by owner. Wire, labels shall include tag numbers as listed on the interconnection diagram.

C. All cables within the relay panels should be run in slotted plastic wire duct. Where wire duct is not provided, cables shall be installed using Thomas and Betts (T&B) nylon ty-wraps or equivalent as approved by owner and anchor blocks.

D. All control wiring shall be verified as to field device and cable integrity before being terminated at either end per the following procedure:

   1). Attach DVM across control cable at access control panel end.

   2). Have person at device end short wires together: verify that continuity is established.

   3). Isolate wires at device and check for continuity from wire to shield, shield to ground, and wire to ground.

   4). If continuity, is observed on any of the readings in step #3, correct problem or replace cable.

   5). If all readings in step #3 show, no continuity, then label and terminate cable at this time.

   6). When terminating field device, strip back shield, cut drain wire and wrap any exposed shielding with black Scotch 33 vinyl tape. Terminate shield drain wire at LNL-1320 assembly.

   7). Record all testing, and submit records to owner.

E. Wires shall be installed in strict accordance with the National Electrical Code, and in particular with Article 725. All communication, signal and instrument cable with insulation of less than 600 volts (Class 2 and 3) shall be kept separate by at least 2” from all power and Class 1 control circuits or a partition shall be installed between these classes of cables.

F. Solderless connections shall be used for wire terminations. Where wires terminate in other than a compression-type terminal, an appropriate T&B Stacon spade-type wire terminator shall be crimped to the wire.

G. Where adjacent terminal blocks are jumpered together as shown on the drawings, standard plug-in jumpers shall be provided and installed by the access control contractor. Wire jumpers shall be used between fused terminal blocks and non-adjacent terminal blocks, and shall have one wire label in the middle where 6" or less, and a wire label on both ends where more than 6”.

H. Any penetration of a fire wall shall be sealed to maintain the fire rating of the wall.

I. Pull boxes shall be a minimum 4 inches square by 2-1/8 inches deep.

J. All boxes shall have a sheet metal cover plate and a nameplate identifying the number for cables in the box. Refer to the interconnection diagrams for tag numbers.

K. All 120 volt power cables shall be installed in separate raceway or enclosures from low voltage cables.

3.5 CONNECTION TO EXISTING CENTRAL ACCESS CONTROL SYSTEM REQUIREMENT

A. Ethernet connection to allow communications to Central Access control System will be by the communications contractor. The access control contractor shall coordinate with this contractor and Owner prior to and while doing this work.

B. During the walk-through of the project, prior to submitting a bid, the access control contractor shall take note of the existing equipment and determine how and where the new access control equipment will be connected. His bid shall include description of how he plans on making the connections as well as the price for doing so. If additional access control equipment, other than described herein, is required to make the connections, the access control contractor shall provide a separate price for this work. If no price is included in his bid for the work, the access control contractor shall be responsible for furnishing all equipment required to make the equipment operable with the central access control system as part of his base bid price.

END OF SECTION